

Final exam, Game Analytics

16/01/2025

1) (1 point)

What are the **most common KPIs (Key Performance Indicators)** used to measure the **economic performance** of a game? Provide a short definition of each

2) (1 point)

Given these two tables:

Players		Sessions	
Player_id	Name	Session_id	Player_id
1	Eva	1	1
2	Elena	2	1
3	Elvira	3	2
4	Esteban		

How many rows would the following query return:

```
Select * FROM Players p LEFT JOIN Sessions s ON p.Player_id = s.Player_id
```

3) (2 points)

You have been hired by LocalThunk as a data analyst for their game Balatro. (If you have not played the game, pick another well-known game). The CEO wants to see **one graph** that provides insights into **game design**. Your job is to decide what **data to collect**, how to **analyze it**, and how to create this graph. You will also explain the steps you'd take to collect and process the data.

Your Tasks:

1. Identify The Key Graph:

Decide which **graph** would provide the most useful insights for improving Balatro's gameplay. Examples:

- A **bar chart** showing the most frequently played cards.
- A **line graph** showing player win rates across different levels or encounters.

2. Steps to Obtain the Data:

Write a clear explanation of the steps you'd take to collect and prepare the data. For example:

- **Data Collection:** What data will you track during the game (e.g., cards played, outcomes of battles, etc.)?
- **Data Organization:** How will you store the data in tables or spreadsheets?
- **Data Analysis:** What calculations or metrics will you use to create the graphs?
- **Data Visualization:** Make a sketch of the graph you would obtain

4) (1 point)

Using the file `sweet.sql` show if the two groups are significantly different in their session length.

5) (4 points)

Our small indie game, **Sweet Break Legend**, was originally designed with **20 levels**. After the game became a major success, we added more content. However, we noticed that **level 20 has a high failure rate**.

To address this, we conducted an **A/B test**, splitting players into two groups:

- **Group A** played an **easier version** of level 20.
- **Group B** played the **original, harder version**.

Using the data in the `sweet.sql` file:

- Should we **keep** the original difficulty or **lower** it for level 20?
- **Explain your reasoning**, backing up your argument with analysis from the data.